Potential climatological impact of Arctic Sea ice retreat: a geometric analysis.

Eloise Davies
HB

Abstract

This essay aims to use the geometry of the Earth's revolution around the Sun to provide insight as to how and when Arctic Sea ice extent changes could affect global albedo. First we look at the relevant angles in Earth, Sun geometry. We look at this in relation to the special case of the Arctic Circle. Then we construct a function that allows us to calculate area of sea ice on an orthogonal plane at all times of year. Incorporating real life data we obtain results for area of sea ice on the orthogonal plane for years 1979-2021 which allows comparative. Extent data is transformed into something relevant to the albedo effect because it takes into account angle of incident solar radiation.

Contents

Introduction 04

Earth and Sun Geometry 04

Area of Ice 07

Incorporating extent data 10

Climatological context 14

Limitations of the model 15

Conclusion 17

References 18

Appendices

Appendix A 19

Introduction

Global warming and climate change are arguably the most consistent news story of this century. As of July 2021, 2016 was the hottest year worldwide and 2011-2020 the hottest decade since records began. (Environmental Protection Agency, 2021) In fact global mean surface temperature has increased at a mean rate of $0.18^{\circ}C$ per decade since 1981. (Lindsey, Dahlman ,2021)

One of the perhaps lesser-known contributing factors to this is reduction in planetary albedo. To explain the concept in short: you will choose to wear a white shirt rather than a black one even in the roaring heat of the British summer. In that decision you are considering your own albedo, as to how much sun you want to reflect and absorb. (Preferably very little of the latter.). The main instance of planetary darkening is Arctic Sea ice retreat. Ice tends to reflect approximately 10x as much solar radiation as sea water. As will be illustrated, there has been unprecedented sea ice retreat in the Arctic, and due to the disparity of the respective albedos this will have played a role in global warming.

However, in order to understand how the impact occurs, we cannot just focus on change in area of Arctic ice. The position of this ice in relation to the Sun changes every day. The angle at which the Sun's rays are incident on the ice will affect the magnitude of the effect of the Sun on actual global surface temperatures. Therefore rather than a number crunching exercise climatological insight transforms into geometry and integrating under approximate functions.

Earth and Sun geometry

The angle, θ

There are two key angles to contend with when it comes to the Earth and the Sun. The first is known as the obliquity. This is defined as the angle that the Earth's axis (a diameter of the Earth which meets the surface at the North and South geographical poles respectively) makes with the perpendicular to the elliptic plane around the Sun. For the purposes of this essay, the obliquity is a constant of 23.5°or $\frac{47}{360}\pi\ rad$. (Williams, 1993). The other which will be defined as θ is the angle between the Earth's axis and the perpendicular to the Sun's rays. This varies with the Earth's position in its revolution around the Sun.

Calculating θ using position in the elliptic would be complicated so some simplifying assumptions have been made. The Earth does not travel in an ellipse around the sun rather a circle that it takes 365 days to traverse the circumference of. Due to these assumptions θ will be a sine function of the day of the year, x (where $x = 0$ is 1st January). To determine this function there are some known values illustrated in Table 1 . The first and third rows are the spring and autumn equinox respectively and the second and fourth the summer and winter solstice.

Table 1 (Jones, (n.d.))

x	θ
78.5	0
171.5	$\frac{47}{360}\pi$
264.5	0
354.5	$-\frac{47}{360}\pi$

With these values an approximate equation for θ as a function of x can be constructed:

$$\theta = \frac{47}{360}\pi \, sin\left(\frac{2\pi}{365}(x - 78.5)\right)$$

The graphing of this equation is illustrated in Figure 1. This gives insights as to how $\frac{d\theta}{dx}$ is at maximum at the equinoxes and minimum at the solstices. One can notice this in real life. Around March and September, the increase and decrease in daylight hours respectively per day is much greater than say around June, where the increments of minutes of sunlight get smaller and smaller.



Figure 1 (Desmos)

So why is this angle, θ relevant? Well, there are two factors affecting the area of Arctic sea ice exposed to the Sun each day of the year. First, there is the actual amount of sea ice which will have a specific surface area. Secondly, the angle at which the Earth's axis is with respect to the Sun's rays.

Figure 2 illustrates the area of the Earth that is illuminated by the Sun at any time during the March and September equinoxes. The Earth's axis is perpendicular to the sun's rays and therefore the day and night will be equal because each geographical place spends half the day in the circle of illumination and the other half out. Figure 3 shows the Earth and Sun at the summer solstice where the Earth's axis is tilted at its max towards the Sun. This means areas in the Northern hemisphere get considerably more time in the Sun than those in the Southern hemisphere. θ is the reason for our seasons as opposed to the popular misconception of distance away from the Sun.

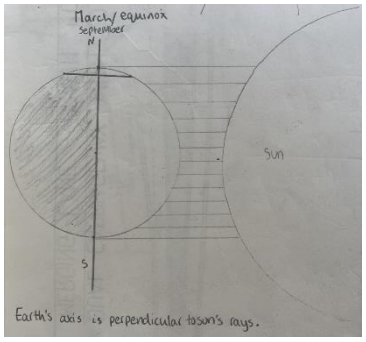


Figure 2

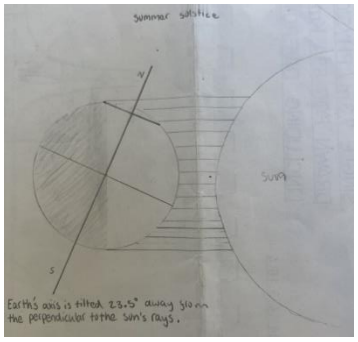


Figure 3

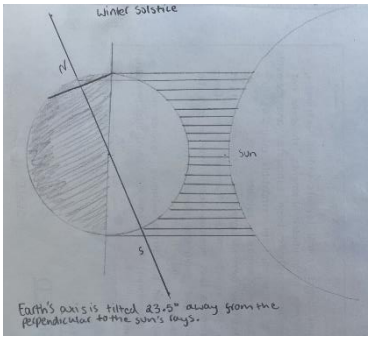


Figure 4

Figure 3 and Figure 4 introduce two very important points.

1) Throughout this essay the assumption is made that the Sun's rays are one dimensional and travel in parallel to the elliptic plane towards the Earth. This is because the Sun is about 150 million km away from the Earth and therefore the photons emitted by the Sun effectively travel in parallel rays. It is the angle of the Earth that controls the intensity of irradiation not the position of the Sun. (Witmer, (n.d.))
2) The Arctic Circle which acts as a reliable base case for the incoming calculations is defined by the position of the Earth at the summer solstice. It is all the area which is encompassed by the 66.5° line of latitude; it is the most southerly line of latitude which has all its points fully illuminated during the summer solstice. This is due to the 23.5° tilt which means the Suns rays can reach the very furthest point of the Arctic circle. Conversely the Arctic Circle is the most southerly line of latitude which is fully in darkness during the winter solstice. The geographic North Pole is at the centre of the Arctic Circle, where all the lines of longitude converge, and experiences 6 months of daylight and 6 months of darkness.

The angle ψ

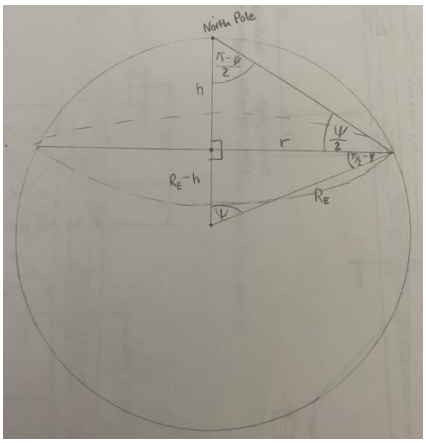


Figure 5

To find the area of ice exposed to solar irradiation on day x, the area is going to be modelled as a spherical cap centred at the geographical North Pole (shown in Figure 5). A spherical cap is defined as "the region of a sphere which lies above a given plane." (Wolfram MathWorld, (n.d.)) Therefore for a given area of ice this can be modelled as the surface area of the spherical cap. From this surface area the unique angle ψ can be found:

$$R_E = radius\ of\ Earth, 6.371 \times 10^6 m$$
$$r = radius\ of\ spherical\ cap, m$$
$$h = height\ of\ spherical\ cap, m$$

$$\psi = angle\ between\ Earth's\ axis\ and\ rim\ of\ spherical\ cap, rad$$
$$S_{cap} = surface\ area\ of\ spherical\ cap, m^2$$

$$S_{cap} = 2\pi R_E h$$
$$h = r tan\frac{\psi}{2}$$
$$r = R_E\ sin\,\psi$$
$$h = tan\frac{\psi}{2} \cdot R_E\ sin\,\psi$$
$$S_{cap} = 2\pi R_E^2\ sin\,\psi\ tan\frac{\psi}{2}$$
$$let\ \psi = 2\phi$$
$$S_{cap} = 2\pi R_E^2 \cdot 2sin\ \phi cos\phi \cdot tan\phi$$
$$= 2\ \pi R_E^2 \cdot 2\sin\phi cos\phi \cdot \frac{sin\phi}{cos\phi}$$
$$= 2\ \pi R_E^2 \cdot 2\ sin^2\ \phi$$
$$= 2\ \pi R_E^2 \cdot 2\left(\frac{1-\cos\psi}{2}\right)$$
$$= 2\ \pi R_E^2 (1-\cos\psi)$$

Area of ice

A framework needs to be found to apply this angle, ψ that has been obtained to calculate area of ice exposed to solar irradiation. Area though is a slight misnomer because as stated in the introduction the angle at which the Sun's rays fall on the Earth will affect the intensity of the irradiation. Therefore instead of worrying how much ice is exposed to the Sun at a certain time of day; it is better to worry about what the Earth will look like resolved onto a plane orthogonal to the Sun's rays. This plane will show all of the Earth that is in the circle of illumination at that one time. This will in effect account for the angle at which the solar radiation is incident because ice that is hit at a less extreme angle will have a greater area on the orthogonal plane.

Ellipses and Segments

So what will the projection look like and is there a pattern? Assuming the Earth is a perfect sphere, it has an infinite number of lines of symmetry therefore its projection onto a plane is going to be a two dimensional shape with an infinite number of lines of symmetry, a circle. Less self-evidently, when a whole spherical cap is visible, its projection will be an ellipse. Figure 6 shows an approximate overview as to how the edge of the Arctic Circle and the North Pole would appear on the orthogonal plane. The North Pole would move down the plane as θ increases until it reaches its max displacement at the summer solstice. The Arctic Circle first appears as a straight line because $\theta = 0$ and therefore the plane and the Earth's axis are orthogonal. Then progressively more of the Arctic Circle is visible on the plane until the final point is illuminated by the Sun and it forms a full ellipse on the plane at the summer solstice.

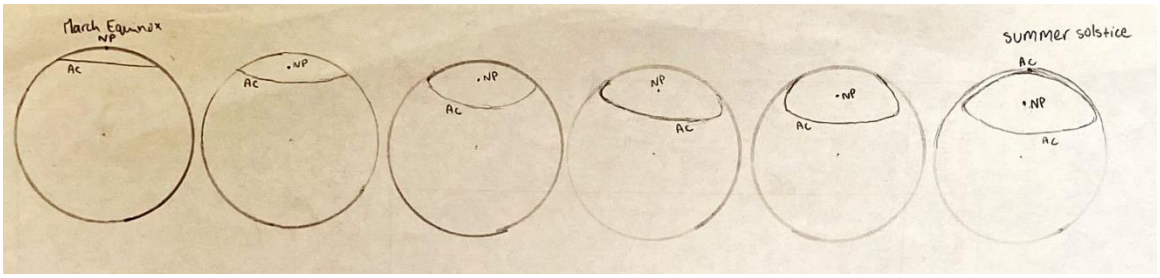


Figure 6

So we know have 3 pieces of information on the appearance of the Arctic Circle on the orthogonal plane:

1) It will not appear on the plane at all during the winter solstice therefore its area is 0.
2) Half of it appears on the plane at the equinoxes as a segment of a circle.
3) It appears on the plane as an ellipse during the summer solstice.

Before we do any calculation let's try to extend this to all spherical caps smaller than the Arctic Circle ie.$\psi < \frac{47}{360}\pi$. Like the Arctic circle they will have no area on the plane at the winter solstice but there will also be a period before and after (length dependent upon the size of ψ) where there will still be no area. Half of the cap will be illuminated at the equinoxes as it is centred at the Arctic Circle. It will appear as a segment just a smaller one. At the summer solstice any size of cap will have its max area on the orthogonal plane however the full cap will be illuminated earlier than the solstice. Like in winter there will be a period before and after however this is not a period of no change. Once the full cap is illuminated an ellipse will form on the plane but the ellipse's area will continue to increase because its dimensions are dependent upon θ.

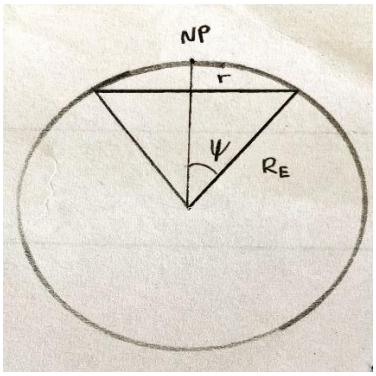


Figure 7

Figure 7 shows a cap of angle ψ with its projection at the equinoxes of a segment. To calculate the area of the segment we can use some two dimensional trigonometry:

$$A = \pi R_E^2 \frac{2\psi}{2\pi} - {}^1\!/_2\, R_E^2 \sin 2\psi$$
$$= R_E^2\psi - {}^1\!/_2\, R_E^2 \sin 2\psi$$
$$= R_E^2\left(\psi - {}^1\!/_2 \sin 2\psi\right)$$

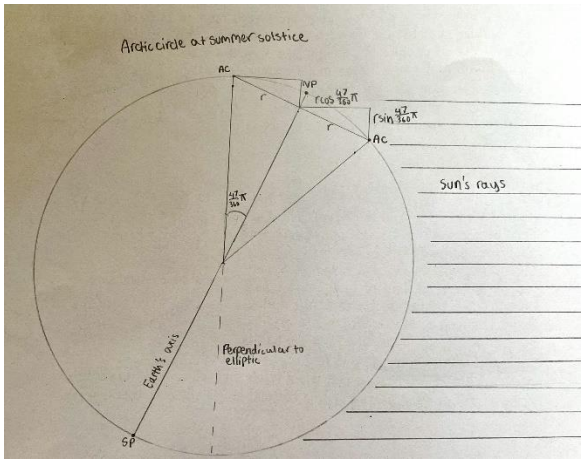


Figure 8

Now to the area of the ellipse… As illustrated in Figure 8 we can draw two triangles right angled triangles with hypotenuse r and angle θ or in this specific case of the Arctic Circle at the summer solstice angle $\frac{47}{360}\pi$. It is easy to conflate the angles θ and ψ particularly in this specific example as they are the same. However it is very important that to calculate the lengths of these triangles we are using θ which is a function of x rather than ψ which is constant for one spherical cap. Figure 9 shows this difference where we have a cap smaller than the Arctic Circle which is fully illuminated before the summer solstice has been reached. On the spherical cap's radii are two triangles which will increase in area when θ increases and the summer solstice is approached.

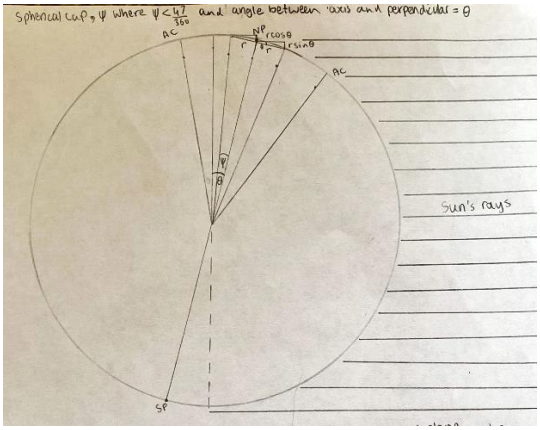


Figure 9

From this we can work out what the dimensions of any ellipse will be on the orthogonal plane. The major axis of the ellipse will be the radius of the spherical cap r which can also be written as:

$$r = R_E \sin\psi$$

The minor axis of the ellipse will be $rsin\theta$. Hence when any spherical cap reaches full illumination we can calculate its area on the plane. Figure 10 shows this general ellipse.

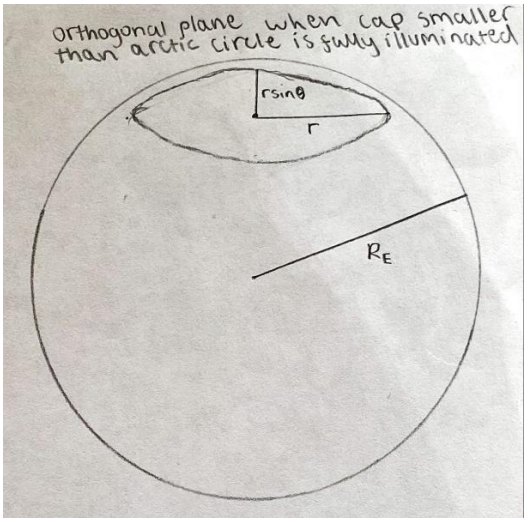


Figure 10

$$ellipse\ area = r^2 \sin\theta\pi$$
$$(using\ r = R_E sin\psi\ when\ \theta = \psi)$$

$$= R_E^2 \sin 3\psi\pi$$

Forming a function

So at first glance if we want to functionalise the area of a spherical cap ψ on the orthogonal plane on day x there's going to have to be a piecewise function involved. The spherical cap has three different sections of behaviour in appearance.

1) There is no area of spherical cap on the orthogonal plane when $\theta \leq -\psi$
2) There is a fully formed ellipse on the orthogonal plane when $\theta \geq \psi$
3) There is an increasing area on the orthogonal plane between no area and a full ellipse when $-\psi < \theta < \psi$ and halfway through this period the area is a segment.

The first two sections are simple to model as the first is a constant therefore just a horizontal line and the second a sine wave dependent on θ. However the last causes more issues…

We could try experimenting with reciprocal and exponential graphs that intersect with our three known points but that would be assuming a relationship that we have not derived. We don't know how the area of spherical cap projected onto the plane actually changes day by day we just know some set points (shown in Table 2). We can instead just connect these points with straight lines that follow the form $y = mx + c$.

Table 2 (These values are found using the function $\theta = \frac{47}{360}\pi\, sin\left(\frac{2\pi}{365}(x - 78.5)\right)$ and the equations for the ellipse and segment.)For the sake of conciseness in the next stage there are attributed letters for the combinations of constants that are used in the next stage!

x	A_{cap}
$\alpha = \frac{365}{2\pi}\left(\arcsin\left(\frac{-360\psi}{47\pi}\right)\right) + 78.5$	0
78.5	$\gamma = R_E^2\left(\psi - {}^1/_2 \sin 2\psi\right)$
$\beta = \frac{365}{2\pi}\left(\arcsin\left(\frac{360\psi}{47\pi}\right)\right) + 78.5$	$\mu = R_E^2 \sin^3\psi\pi$
$339.5 - \beta = 261 - \frac{365}{2\pi}\left(\arcsin\left(\frac{360\psi}{47\pi}\right)\right)$	$\mu = R_E^2 \sin^3\psi\pi$
261	$\gamma = R_E^2\left(\psi - {}^1/_2 \sin 2\psi\right)$

$339.5 - \alpha = 261 - \frac{365}{2\pi}\left(\arcsin\left(\frac{-360\psi}{47\pi}\right)\right)$	0

The following piece wise function is given mostly in terms of x and ψ (which will be a constant when the function is used). θ is used for some of the boundaries as it makes more intuitive sense and saves space. So keep in mind that $\theta = \frac{47}{360}\pi \sin\left(\frac{2\pi}{365}(x - 78.5)\right)$. The letters for the long constant terms introduced in *Table 2* are also used.

$$A_{cap} = \begin{cases} 0 & \theta \leq -\psi \\ \frac{\gamma}{78.5 - \alpha}x + \frac{\gamma\alpha}{\alpha - 78.5} & -\psi < \theta \leq 0 \; for \; x \leq 78.5 \\ \frac{\mu - \gamma}{\beta - 78.5}x + \left(\frac{78.5(\gamma - \mu)}{\beta - 78.5} + \gamma\right) & 0 < \theta \leq \psi \; for \; x \leq 169.75 \\ R_E^2 \sin^2\psi \sin\left(\frac{47}{360}\pi \sin\left(\frac{2\pi}{365}(x - 78.5)\right)\right)\pi & \theta > \psi \\ \frac{\mu - \gamma}{78.5 - \beta}x + \left(\frac{261(\gamma - \mu)}{78.5 - \beta} + \gamma\right) & 0 < \theta \leq \psi \; for \; x > 169.75 \\ \frac{\gamma}{\alpha - 78.5}x + \left(\frac{\gamma\alpha}{\alpha - 78.5} - \frac{339.5}{\alpha - 78.5}\right) & -\psi < \theta \leq 0 \; for \; x > 78.5 \end{cases}$$

Incorporating extent data

We have a function that allows us to find the area of the spherical cap size ψ on the plane on day x (A_{cap}) and a formula that allows us to convert a piece of area data into an angle ψ, $S_{cap} = 2\,\pi R_E^2(1 - \cos\psi)$. Therefore we can now introduce the data into the 'equation'.

The satellite era began in the early 1970s. The National Snow and Ice Data Centre began to collect useful satellite data concerning sea ice in late 1978 through NASA's launch of the Scanning Multichannel Microwave Radiometer (SSMR) satellite. This passive microwave data is available all year round as the sensor can record data during the six months of Arctic darkness and it can also penetrate through clouds. However it does have some limitations. It may show ice where none exists due to signal variation between land and water along coasts or atmospheric interference from raid or high winds. It also may not detect ice because of it being thin and newly formed. Yet these limitations are consistent throughout the recorded years. Therefore when comparing year by rear these errors do not affect the comparison. (NSIDC, 2012)

Although the word 'area' of ice has been used throughout the essay so far. We will be using the measure of ice extent from the National Snow and Ice Data Centre data set Sea Ice Index Version 3 (Fetterer et al, 2017). This is because satellites tent to measure extent more accurately than other measurements like thickness. Extent classifies regions as having two distinct states ice covered or not ice covered with the threshold being approximately 15%. This is in comparison to the measurement of area which measures ice in individual pixels. The decision to look at extent over area is particularly impactful during the summer months as to satellite sensors the surface melt (which is water on top of sea ice) appears to be open water which would mean the amount of ice is underestimated. The sea ice in the context of albedo is its most important in Summer as the rays are hitting it at a less extreme angle.

The Sea Ice Index Version 3 (Fetterer et al, 2017) begins in 1979. We will be using monthly data rather than daily data as most doing climatological studies do. It is subject to less

random variation ,due circumstances like weather, than daily ice extent so its therefore better for analysing long term trends.

(See Appendix A) We have the monthly extent in m^2 and from that we calculate ψ. The result we to compute is the monthly total of ice on the orthogonal plane. This can be calculated by integrating with respect to x under the relevant (one or two) sub-functions for which our limits of x apply. Our scale of x begins with $x = 0$ as the 1st of January. Therefore our limits of the months will be as follows (see *Table 3*). There appears to be "overlap" but these are the boundaries of the month that we still have to make sure we integrate under.

Table 3

Month	x limits
January	$[0,31]$
February	$[31,59]$
March	$[59,90]$
April	$[90,120]$
May	$[120,151]$
June	$[151,181]$
July	$[181,212]$
August	$[212,243]$
September	$[243,273]$
October	$[273,304]$
November	$[304,334]$
December	$[334,365]$

Calculating the integrals for some of these months will be easier than others. For example in December every year has a value of ψ small enough meaning the whole spherical cap does not appear on the orthogonal plane at all meaning the monthly total is 0. This also occurs for some of the November data in later years – already illustrating the out of the norm decline in Arctic Sea ice. Then for January, February, March, April, September, October and November we solely integrate under linear functions. Below is an example of February in 1979:

$$\psi = 0.3581$$

$$relevant\ sub\ function:\ A_{cap} = \frac{\gamma}{78.5-\alpha}x + \frac{\gamma\alpha}{\alpha - 78.5}$$

$$A_{February\ 1979} = \int_{31}^{59} \frac{R_E^2\left(\psi - {}^1/_2 \sin 2\psi\right)}{-\frac{365}{2\pi}\left(\arcsin\left(\frac{-360\psi}{47\pi}\right)\right)}x + \frac{\left(R_E^2\left(\psi - {}^1/_2 \sin 2\psi\right)\right)\cdot\left(\frac{365}{2\pi}\left(\arcsin\left(\frac{-360\psi}{47\pi}\right)\right)+78.5\right)}{\frac{365}{2\pi}\left(\arcsin\left(\frac{-360\psi}{47\pi}\right)\right)}\,dx$$

$$= \left[\frac{R_E^2\left(\psi - {}^1/_2 \sin 2\psi\right)}{-2\cdot\frac{365}{2\pi}\left(\arcsin\left(\frac{-360\psi}{47\pi}\right)\right)}x^2 + \left(\frac{\left(R_E^2\left(\psi - {}^1/_2 \sin 2\psi\right)\right)\cdot\left(\frac{365}{2\pi}\left(\arcsin\left(\frac{-360\psi}{47\pi}\right)\right)+78.5\right)}{\frac{365}{2\pi}\left(\arcsin\left(\frac{-360\psi}{47\pi}\right)\right)}\right)x\right]_{31}^{59}$$

(from Appendix A)

$$R_E^2\left(\psi - {}^1/_2 \sin 2\psi\right) = 1.211 \times 10^{12}$$

$$\frac{365}{2\pi}\left(\arcsin\left(\frac{-360\psi}{47\pi}\right)\right) = -61.67$$

$$-2\cdot\frac{365}{2\pi}\left(\arcsin\left(\frac{-360\psi}{47\pi}\right)\right) = 123.3$$

$$\left(\frac{365}{2\pi}\left(\arcsin\left(\frac{-360\psi}{47\pi}\right)\right)+78.5\right) = 16.83$$

$$A_{February\ 1979} = \left(\frac{1.21135 \times 10^{12}}{123.3} \cdot 59^2 + \frac{1.21135 \times 10^{12} \cdot 16.83}{-61.67} \cdot 59\right) - \left(\frac{1.211 \times 10^{12}}{123.3} \cdot 31^2 + \frac{1.211 \times 10^{12} \cdot 16.83}{-61.67} \cdot 31\right) = 1.55 \times 10^{13} m^2$$

The result $1.55 \times 10^{13} m^2$ to 3 significant figures is the same as the spreadsheet's $1.54933 \times 10^{13} m^2$. The process for the rest of the linear months is broadly the same. With some the months are split between the subfunction for before the equinox and after the equinox.

Where the computation differs is once the size of ψ and the month being calculated means the whole spherical cap is visible on the orthogonal plane. The area then follows the function $R_E^2 \sin^2\psi \sin\left(\frac{47}{360}\pi\, sin\left(\frac{2\pi}{365}(x-78.5)\right)\right)\pi$ and this is difficult to integrate with respect to x. We will use an example of July 1979 to show how instead we convert the limits into θ limits and integrate by substitution of θ.

$$\psi = 0.2853$$

$$A_{July\ 1979} = \int_{181}^{212} R_E^2 \sin^2\psi \sin\left(\frac{47}{360}\pi\, sin\left(\frac{2\pi}{365}(x-78.5)\right)\right)\pi\, dx$$

$$\theta = \frac{47}{360}\pi\, sin\left(\frac{2\pi}{365}(x-78.5)\right)$$

$$1 = \frac{2\pi}{365} \cdot \frac{47}{360}\pi\, cos\left(\frac{2\pi}{365}(x-78.5)\right) \cdot \frac{dx}{d\theta}$$

$$\left(\frac{47}{360}\pi\right)^2 \left(\sin^2\left(\frac{2\pi}{365}(x-78.5)\right) + \cos^2\left(\frac{2\pi}{365}(x-78.5)\right)\right) = \left(\frac{47}{360}\pi\right)^2$$

$$\left(\frac{47}{360}\pi\right)^2 \left(\left(\frac{360}{47\pi}\theta\right)^2 + \cos^2\left(\frac{2\pi}{365}(x-78.5)\right)\right) = \left(\frac{47}{360}\pi\right)^2$$

$$cos\left(\frac{2\pi}{365}(x-78.5)\right) = \sqrt{1-\left(\frac{360}{47\pi}\theta\right)^2}$$

$$1 = \frac{47}{65700}\pi^2 \cdot \sqrt{1-\left(\frac{360}{47\pi}\theta\right)^2} \cdot \frac{dx}{d\theta}$$

$$\frac{dx}{d\theta} = \frac{65700}{47\pi^2 \cdot \sqrt{1-\left(\frac{360}{47\pi}\theta\right)^2}}$$

$$A_{July\ 1979} = \int_{0.3064}^{0.4025} R_E^2 \sin^2\psi\pi \frac{65700\sin(\theta)}{47\pi^2 \cdot \sqrt{1-\left(\frac{360}{47\pi}\theta\right)^2}} d\theta$$

$$= \int_{0.3064}^{0.4025} R_E^2 \sin^2\psi\pi \frac{\sin(\theta)}{\frac{47\pi^2}{65700}\cdot\frac{360}{47\pi}\sqrt{\left(\frac{47\pi}{360}\right)^2-\theta^2}}d\theta$$

$$= R_E^2 \sin^2\psi\pi \int_{0.3064}^{0.4025} \frac{sin\theta}{\left(\frac{2\pi}{365}\right)\sqrt{\frac{2209}{129600}\pi^2-\theta^2}}\,d\theta$$

$$= \frac{365}{2\pi} R_E^2 \sin^2\psi\pi \int_{0.3064}^{0.4025} \frac{sin\theta}{\sqrt{\frac{2209}{129600}\pi^2-\theta^2}}\,d\theta$$

$$= \frac{365}{2} R_E^2 \sin^2\psi \int_{0.3064}^{0.4025} \frac{sin\theta}{\sqrt{\frac{2209}{129600}\pi^2-\theta^2}}\,d\theta$$

$$A_{July\ 1979} = 5.868 \times 10^{14} \cdot 0.1895$$
$$= 1.11 \times 10^{14} m^2$$

The result $1.11 \times 10^{14} m^2$ to 3 significant figures is the same as the spreadsheet's $1.112 \times 10^{14} m^2$.

There isn't a known integral that can be written algebraically in this circumstance. Therefore we have to use numerical integration to compute the answer. May and August require a significant amount of computation because the limits (functions of ψ) had to be changed for each integration and they did not only use the sine function but the linear functions as well. To put this into context of the spherical cap… May and August have in their end and beginning respectively the whole spherical cap illuminated. The rest of the month the projection onto the plane is modelled by the linear functions where not all the spherical cap is illuminated.

Appendix A shows all the computed monthly integral values as well as the yearly totals which we will be using.

Climatological context

The albedo effect and its resulting feedback loops result in some of the largest uncertainties on current climate models (Perkins, 2019) and no where are these more prevalent than in the Arctic. *Figure 11* illustrates the results obtained from the geometric analysis of the National Snow and Ice Data Centre's Sea Ice Index. From the average across the first 10 years that data was recorded (excluding 1988 and instead including 1990 due to gaps in data) to the year's area on the orthogonal plane in 2021 there is an astonishing 16% decline. (Appendix A) This means because crucially we already took into account the angle of the Sun's rays with the invocation of the orthogonal plane there will be a deficit of albedo effect of 16% in 2021 compared to the 1980s.

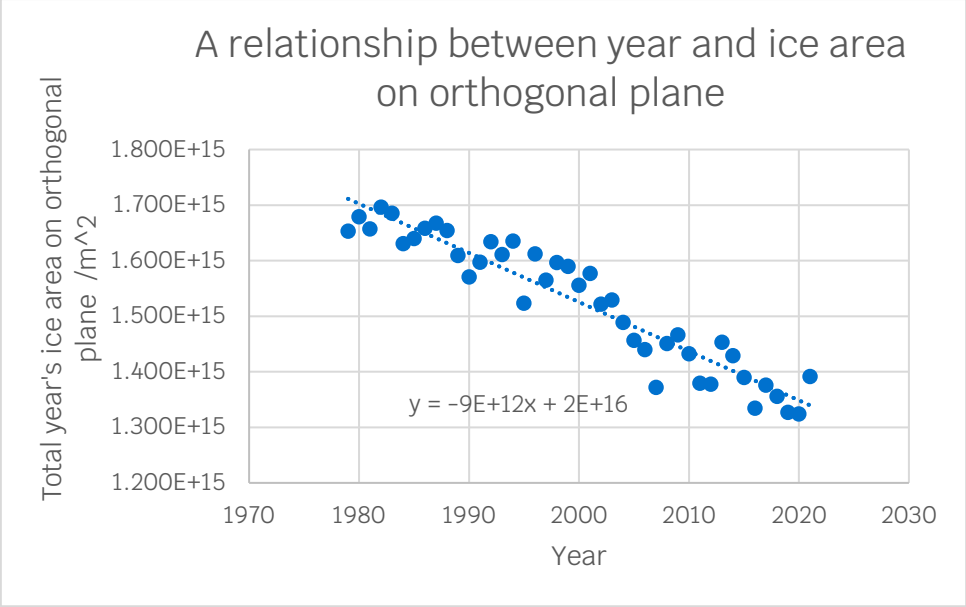


Figure 11 (Appendix A)

Albedo is measured as the fraction of the solar radiation that is reflected back by the material. Ocean albedo is approximately 0.06 whereas bare sea ice varies between 0.5-0.7. Snow has an even higher albedo at around 0.9 whereas melt pools' varies between 0.4 and 0.5. If we make the simplifying assumption that the ice on our orthogonal plane is ice with an albedo between 0.5-0.7 and where there is not ice but used to be in previous years is ocean with an albedo of 0.06 we can gauge how much effect this Arctic ice retreat has had on Arctic albedo. (NSIDC, 2020)

The whole orthogonal plane's represents the total solar energy incident on the Earth scaled accordingly. We can calculate the total orthogonal plane area across a whole year which in a way represents and unachievable "maxima" for the ice.

$$A_{op} = \pi R_E^2$$
$$= 1.4 \times 10^{14}$$
$$365\ A_{op} = 5.195 \times 10^{16} m^2$$
$$A_{1979} = 1.653 \times 10^{15} m^2$$
$$\%A_{1979} = \frac{1.653 \times 10^{15}}{5.195 \times 10^{16}} \cdot 100\% = 3.18\%$$
$$A_{2021} = 1.391 \times 10^{15} m^2$$
$$\%A_{2021} = \frac{1.391 \times 10^{15}}{5.195 \times 10^{16}} \cdot 100\% = 2.68\%$$

So instead of having 0.0318 proportion of the Earth receiving sunlight reflecting 50-70% of solar radiation back into the atmosphere we, in 2021, have 0.0268 reflecting 50-70% and the remaining 0.005 reflecting 6% back. Less reflection of solar radiation means radiative forcing, in effect, the Earth heats up. To quantify the level that Arctic ice retreat has contributed to global radiative forcing is difficult with our model.

However looking at *Figure 11* we can yield some dramatic results. The line of best fit has the linear equation:

$$y = -9 \times 10^{12} x + 2 \times 10^{16}$$

If we solve for x when $y = 0$ we find there will be no ice on the orthogonal plane throughout the whole year in year 2222. This is of course assuming the rate of change continues throughout the next 200 years which it will not as it will accelerate. But, why is this result so interesting and again why is the orthogonal plane so important? The model by no means suggests that there will be no ice at all in the Arctic Sea in 2222 that would be the case if we were linearising the extent data. It means there will be no ice actually reflecting the solar radiation which is the more interesting part. It doesn't matter to the climate in the sense of the albedo effect if there is ice that spends its whole lifetime in darkness. This is why the so-called "blue ocean event"(a day where there is no Arctic Sea ice) is so worrying because it will happen in September or August when it is very sunny in the Arctic. (ClimateTippingPoints, 2019)

Limitations of the model

The geometric model yielded lots of results with little attention to real life circumstance other than Earth Sun geometry , which is pretty constant, and the NSIDC sea ice index. But some of the helpful approximations could have led to errors.

The Earth revolves around the Sun in an ellipse shape and also changes in speed at some points during its journey around the Sun . The angle θ more specifically depends upon the Earth's position in its revolution but using days is a simplifying assumption that went on to serve a purpose for modelling the area on the orthogonal plane and when computing the integrals to decide which subfunction to use.

The Arctic Sea ice is not symmetric and reaches more extreme latitudes in some areas than others which is illustrated in *Figure 12*. Therefore the approximations for the area on the orthogonal plane would be underestimations as ice at lower latitudes would be hit by the Sun at a less extreme angle would experience greater solar irradiation.

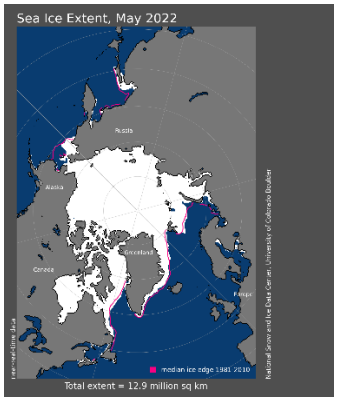


Figure 12

We used linear subfunctions to model the behaviour of the area on the orthogonal plane between no illumination and the whole spherical cap being illuminated. This would lead to error in the monthly integrals as the relationship would be more curved and complex. However to some extent this error was mitigated as we calculated a certain point at the equinoxes meaning there were four linear functions as opposed to two.

To further the model, it would be interesting to extend it to calculating the Arctic ice retreat's contribution to global radiative forcing. This has been calculated by other people with very varying results. Some argue negative cloud feedbacks mean the ice loss has no impact on radiative forcing whilst others just differ on the magnitude of the radiative forcing.

Conclusion

In analysing real life data we have provided the framework for an understanding of how Arctic Sea Ice retreat affects the Earth's ability to reflect solar radiation. The essays key focus has been to model how this decreasing ice extent behaves in relation to where the sun is incident because ice that's in darkness doesn't do anything to albedo! Just by relying on figures for the obliquity of the Earth, radius of the Earth and the knowledge of the days and months in a year we have transformed an ice extent data set to something relevant for albedo effect climatological analysis. Along the way we computed the approximate figure that if ice retreat continues as it is we will have non-reflecting Arctic Sea ice by 2222. There could have been forecasting data and further scientific understanding of Sea ice retreat and its behaviours incorporated into the model but the elementary geometric model has sufficed for some interesting revelations. It is inextricable that global temperatures are on the rise due to Arctic Sea ice retreat.

References

Climatetippingpoints.info. 2019. “Fact-Check: will an ice-free Arctic trigger a climate catastrophe?”2021. (Accessed 28th June 2022) https://climatetippingpoints.info/2019/04/02/fact-check-will-an-ice-free-arctic-trigger-a-climate-catastrophe/

Environmental Protection Agency. (n.d.) "Climate Change Indicators: U.S. and Global Temperature." 2021. (Accessed 25th June 2022). https://www.epa.gov/climate-change-indicators-us-and-global-temperature

Fetterer, F.,K. Knowles, W. N. Meier, M. Savoie, and A. K. Windnagel. 2017, updated daily. Sea Ice Index, Version 3. Sea Ice Concentrations from Nimbus-7 SMMR and DMSP SSM/I-SSMIS Passive Microwave data. Boulder, Colorado USA. NSIDC: National Snow and Ice Data Centre. (Accessed 24th June 2022) https://doi.org/10.7265/N5K072F8.

Jones, G. (n.d.) "What is the Equinox? And What is the Solstice?". (Accessed 25th June 2022) https://www.timeanddate.com/astronomy/equinox-solstice.html

Lindsey, R, Dahlman, L. 2021. "Climate Change: Global Temperature". 2021. (Accessed 25th June 2022). https://www.climate.gov/news-features/understanding-climate/climate-change-global-temperature

National Snow and Ice Data Centre. 2012. “Sometimes readers report that our maps show ice incorrectly , compared to on-the-ground observations or other data sources. Why is this?”. (Accessed 20th June 2022) https://www.nsidc.org/arcticseaicenews/faq/#area-extent

National Snow and Ice Data Centre. 2020. “Thermodynamics: Albedo”. (Accessed 28th June 2022) https://nsidc.org/cryosphere/seaice/processes/albedo.html

Perkins, S. 2019. “Albedo is a simple concept that plays complicated roles in climate and astronomy”. 2019. (Accessed 28th June 2022) https://www.pnas.org/doi/10.1073/pnas.1918770116

Williams, G. 1993. "History of the earth's obliquity". Earth Science Reviews volume 34, issue 1: pages 1-45. https://doi.org/10.1016/0012-8252(93)90004-Q

Witmer, L. (n.d.) "1.3. Earth's Tilted Axis and the Seasons". (Accessed 26th June 2022). https://www.e-education.psu.edu/eme811/node/642

Wolfram MathWorld. (n.d.) "Spherical Cap". (Accessed 26th June 2022) https://www.mathworld.wolfram.com/SphericalCap.html

Appendix A

	January			February			March			April		
year	extent /m^2	ψ	monthly total of ice on orthogonal plane /m^2	extent /m^2	ψ	monthly total of ice on orthogonal plane /m^2	extent/m^2	ψ	monthly total of ice on orthogonal plane /m^2	extent/m^2	ψ	monthly total of ice on orthogonal plane /m^2
1979	1.541E+13	0.3494	1.095E+12	1.618E+13	0.3581	1.549E+13	1.634E+13	0.3599	5.104E+14	1.545E+13	0.3499	3.259E+14
1980	1.486E+13	0.3431	7.593E+11	1.596E+13	0.3557	1.497E+13	1.604E+13	0.3566	5.042E+14	1.543E+13	0.3496	3.256E+14
1981	1.491E+13	0.3436	7.867E+11	1.56E+13	0.3516	1.413E+13	1.563E+13	0.3519	4.953E+14	1.501E+13	0.3448	3.191E+14
1982	1.518E+13	0.3468	9.451E+11	1.597E+13	0.3558	1.499E+13	1.604E+13	0.3566	5.042E+14	1.547E+13	0.3501	3.262E+14
1983	1.494E+13	0.3440	8.034E+11	1.601E+13	0.3562	1.509E+13	1.609E+13	0.3571	5.052E+14	1.517E+13	0.3466	3.216E+14
1984	1.447E+13	0.3385	5.655E+11	1.53E+13	0.3481	1.345E+13	1.558E+13	0.3513	4.942E+14	1.501E+13	0.3448	3.191E+14
1985	1.472E+13	0.3414	6.857E+11	1.547E+13	0.3501	1.384E+13	1.589E+13	0.3549	5.010E+14	1.536E+13	0.3488	3.245E+14
1986	1.489E+13	0.3434	7.757E+11	1.578E+13	0.3536	1.455E+13	1.591E+13	0.3551	5.014E+14	1.506E+13	0.3454	3.199E+14
1987	1.497E+13	0.3443	8.203E+11	1.605E+13	0.3567	1.518E+13	1.582E+13	0.3541	4.995E+14	1.521E+13	0.3471	3.222E+14
1988	-9.999E+15			1.558E+13	0.3513	1.409E+13	1.596E+13	0.3557	5.025E+14	1.512E+13	0.3461	3.208E+14
1989	1.495E+13	0.3441	8.090E+11	1.55E+13	0.3504	1.391E+13	1.542E+13	0.3495	4.907E+14	1.433E+13	0.3368	3.080E+14
1990	1.478E+13	0.3421	7.167E+11	1.558E+13	0.3513	1.409E+13	1.587E+13	0.3546	5.006E+14	1.465E+13	0.3406	3.133E+14
1991	1.436E+13	0.3372	5.169E+11	1.525E+13	0.3476	1.334E+13	1.542E+13	0.3495	4.907E+14	1.486E+13	0.3431	3.167E+14
1992	1.464E+13	0.3405	6.457E+11	1.546E+13	0.3500	1.381E+13	1.548E+13	0.3502	4.920E+14	1.468E+13	0.3409	3.138E+14
1993	1.49E+13	0.3435	7.812E+11	1.569E+13	0.3526	1.434E+13	1.581E+13	0.3540	4.993E+14	1.508E+13	0.3456	3.202E+14
1994	1.473E+13	0.3415	6.908E+11	1.556E+13	0.3511	1.404E+13	1.555E+13	0.3510	4.936E+14	1.489E+13	0.3434	3.172E+14
1995	1.459E+13	0.3399	6.215E+11	1.523E+13	0.3473	1.330E+13	1.526E+13	0.3477	4.871E+14	1.445E+13	0.3382	3.100E+14
1996	1.418E+13	0.3350	4.430E+11	1.515E+13	0.3464	1.312E+13	1.512E+13	0.3461	4.839E+14	1.422E+13	0.3355	3.062E+14
1997	1.442E+13	0.3379	5.431E+11	1.544E+13	0.3497	1.377E+13	1.547E+13	0.3501	4.918E+14	1.456E+13	0.3395	3.118E+14
1998	1.472E+13	0.3414	6.857E+11	1.575E+13	0.3533	1.448E+13	1.56E+13	0.3516	4.947E+14	1.489E+13	0.3434	3.172E+14
1999	1.436E+13	0.3372	5.169E+11	1.531E+13	0.3483	1.348E+13	1.54E+13	0.3493	4.902E+14	1.508E+13	0.3456	3.202E+14
2000	1.422E+13	0.3355	4.588E+11	1.514E+13	0.3463	1.310E+13	1.522E+13	0.3472	4.862E+14	1.456E+13	0.3395	3.118E+14
2001	1.42E+13	0.3353	4.509E+11	1.521E+13	0.3471	1.325E+13	1.552E+13	0.3507	4.929E+14	1.486E+13	0.3431	3.167E+14
2002	1.427E+13	0.3361	4.791E+11	1.534E+13	0.3486	1.354E+13	1.535E+13	0.3487	4.891E+14	1.43E+13	0.3365	3.075E+14
2003	1.439E+13	0.3375	5.299E+11	1.519E+13	0.3469	1.321E+13	1.548E+13	0.3502	4.920E+14	1.451E+13	0.3389	3.110E+14
2004	1.403E+13	0.3332	3.864E+11	1.491E+13	0.3436	1.259E+13	1.499E+13	0.3446	4.809E+14	1.399E+13	0.3328	3.023E+14
2005	1.366E+13	0.3288	2.657E+11	1.437E+13	0.3373	1.145E+13	1.469E+13	0.3411	4.738E+14	1.409E+13	0.3340	3.040E+14
2006	1.347E+13	0.3265	2.135E+11	1.432E+13	0.3367	1.134E+13	1.442E+13	0.3379	4.673E+14	1.391E+13	0.3318	3.010E+14
2007	1.37E+13	0.3293	2.775E+11	1.451E+13	0.3389	1.174E+13	1.454E+13	0.3393	4.702E+14	1.385E+13	0.3311	3.000E+14
2008	1.389E+13	0.3316	3.377E+11	1.495E+13	0.3441	1.268E+13	1.518E+13	0.3468	4.852E+14	1.435E+13	0.3371	3.084E+14
2009	1.391E+13	0.3318	3.444E+11	1.481E+13	0.3425	1.238E+13	1.498E+13	0.3444	4.806E+14	1.45E+13	0.3388	3.108E+14
2010	1.374E+13	0.3297	2.896E+11	1.458E+13	0.3398	1.189E+13	1.514E+13	0.3463	4.843E+14	1.466E+13	0.3407	3.135E+14
2011	1.346E+13	0.3263	2.109E+11	1.436E+13	0.3372	1.142E+13	1.455E+13	0.3394	4.704E+14	1.411E+13	0.3342	3.044E+14
2012	1.373E+13	0.3296	2.865E+11	1.455E+13	0.3394	1.182E+13	1.52E+13	0.3470	4.857E+14	1.463E+13	0.3404	3.130E+14
2013	1.37E+13	0.3293	2.775E+11	1.472E+13	0.3414	1.218E+13	1.503E+13	0.3450	4.818E+14	1.43E+13	0.3365	3.075E+14
2014	1.365E+13	0.3287	2.628E+11	1.442E+13	0.3379	1.155E+13	1.476E+13	0.3419	4.755E+14	1.409E+13	0.3340	3.040E+14
2015	1.36E+13	0.3280	2.485E+11	1.44E+13	0.3376	1.151E+13	1.437E+13	0.3373	4.661E+14	1.389E+13	0.3316	3.006E+14
2016	1.346E+13	0.3263	2.109E+11	1.42E+13	0.3353	1.109E+13	1.44E+13	0.3376	4.668E+14	1.368E+13	0.3290	2.970E+14
2017	1.319E+13	0.3230	1.481E+11	1.412E+13	0.3343	1.093E+13	1.429E+13	0.3363	4.641E+14	1.375E+13	0.3299	2.982E+14
2018	1.308E+13	0.3217	1.260E+11	1.397E+13	0.3325	1.063E+13	1.43E+13	0.3365	4.644E+14	1.37E+13	0.3293	2.974E+14
2019	1.357E+13	0.3277	2.402E+11	1.439E+13	0.3375	1.149E+13	1.457E+13	0.3397	4.709E+14	1.343E+13	0.3260	2.927E+14
2020	1.364E+13	0.3285	2.599E+11	1.464E+13	0.3405	1.201E+13	1.473E+13	0.3415	4.748E+14	1.362E+13	0.3283	2.960E+14
2021	1.35E+13	0.3268	2.213E+11	1.439E+13	0.3375	1.149E+13	1.466E+13	0.3407	4.731E+14	1.379E+13	0.3304	2.989E+14
2022	1.388E+13	0.3314	3.343E+11	1.461E+13	0.3401	1.195E+13	1.459E+13	0.3399	4.714E+14	1.404E+13	0.3334	3.032E+14

	May			June			July			August		
year	extent/m^2	ψ	monthly total of ice on orthogonal plane /m^2	extent/m^2	ψ	monthly total of ice on orthogonal plane /m^2	extent/m^2	ψ	monthly total of ice on orthogonal plane /m^2	extent/m^2	ψ	monthly total of ice on orthogonal plane /m^2
1979	1.386E+13	0.3312	2.345E+14	1.253E+13	0.3148	1.424E+14	1.031E+13	0.2853	1.112E+14	8.04E+12	0.2518	1.154E+14
1980	1.379E+13	0.3304	2.324E+14	1.22E+13	0.3106	1.387E+14	1.01E+13	0.2824	1.090E+14	7.98E+12	0.2508	1.140E+14
1981	1.38E+13	0.3305	2.324E+14	1.243E+13	0.3135	1.412E+14	1.027E+13	0.2848	1.108E+14	7.84E+12	0.2486	1.110E+14
1982	1.397E+13	0.3325	2.384E+14	1.248E+13	0.3141	1.418E+14	1.037E+13	0.2861	1.118E+14	8.14E+12	0.2533	1.176E+14
1983	1.349E+13	0.3267	2.203E+14	1.23E+13	0.3118	1.398E+14	1.057E+13	0.2889	1.139E+14	8.19E+12	0.2541	1.187E+14
1984	1.358E+13	0.3278	2.246E+14	1.215E+13	0.3099	1.381E+14	9.98E+12	0.2807	1.077E+14	7.77E+12	0.2475	1.095E+14
1985	1.407E+13	0.3337	2.421E+14	1.222E+13	0.3108	1.389E+14	9.74E+12	0.2773	1.052E+14	7.4E+12	0.2415	1.016E+14
1986	1.338E+13	0.3254	2.176E+14	1.198E+13	0.3077	1.363E+14	1.015E+13	0.2831	1.095E+14	7.98E+12	0.2508	1.140E+14
1987	1.374E+13	0.3297	2.302E+14	1.249E+13	0.3143	1.419E+14	1.033E+13	0.2856	1.114E+14	7.63E+12	0.2452	1.065E+14
1988	1.356E+13	0.3276	2.239E+14	1.194E+13	0.3072	1.358E+14	9.81E+12	0.2783	1.059E+14	7.89E+12	0.2494	1.121E+14
1989	1.304E+13	0.3212	2.060E+14	1.224E+13	0.3111	1.391E+14	1.013E+13	0.2828	1.093E+14	7.88E+12	0.2492	1.119E+14
1990	1.323E+13	0.3235	2.124E+14	1.164E+13	0.3033	1.325E+14	9.25E+12	0.2702	9.998E+13	6.8E+12	0.2314	8.924E+13
1991	1.347E+13	0.3265	2.207E+14	1.211E+13	0.3094	1.377E+14	9.51E+12	0.2739	1.027E+14	7.42E+12	0.2418	1.020E+14
1992	1.322E+13	0.3234	2.121E+14	1.215E+13	0.3099	1.381E+14	1.032E+13	0.2855	1.113E+14	7.93E+12	0.2500	1.129E+14
1993	1.337E+13	0.3252	2.173E+14	1.187E+13	0.3063	1.350E+14	9.48E+12	0.2735	1.024E+14	7.33E+12	0.2403	1.001E+14
1994	1.362E+13	0.3283	2.260E+14	1.202E+13	0.3082	1.367E+14	9.93E+12	0.2800	1.072E+14	7.64E+12	0.2454	1.067E+14
1995	1.297E+13	0.3203	2.036E+14	1.144E+13	0.3007	1.303E+14	8.99E+12	0.2663	9.722E+13	6.74E+12	0.2304	8.804E+13
1996	1.309E+13	0.3218	2.076E+14	1.208E+13	0.3090	1.374E+14	1.016E+13	0.2832	1.096E+14	8.18E+12	0.2540	1.185E+14
1997	1.318E+13	0.3229	2.107E+14	1.174E+13	0.3046	1.336E+14	9.41E+12	0.2725	1.017E+14	7.29E+12	0.2397	9.928E+13
1998	1.359E+13	0.3279	2.249E+14	1.171E+13	0.3042	1.333E+14	9.42E+12	0.2726	1.018E+14	7.51E+12	0.2433	1.039E+14
1999	1.375E+13	0.3299	2.306E+14	1.178E+13	0.3051	1.340E+14	9.49E+12	0.2737	1.025E+14	7.23E+12	0.2387	9.803E+13
2000	1.315E+13	0.3225	2.097E+14	1.167E+13	0.3037	1.328E+14	9.51E+12	0.2739	1.027E+14	7.17E+12	0.2377	9.679E+13
2001	1.351E+13	0.3270	2.221E+14	1.146E+13	0.3009	1.305E+14	9.07E+12	0.2675	9.807E+13	7.46E+12	0.2425	1.029E+14
2002	1.297E+13	0.3203	2.036E+14	1.158E+13	0.3025	1.318E+14	9.27E+12	0.2704	1.002E+14	6.6E+12	0.2280	8.526E+13
2003	1.302E+13	0.3209	2.053E+14	1.16E+13	0.3028	1.320E+14	9.21E+12	0.2696	9.955E+13	6.94E+12	0.2338	9.207E+13
2004	1.256E+13	0.3151	1.900E+14	1.145E+13	0.3008	1.304E+14	9.43E+12	0.2728	1.019E+14	6.85E+12	0.2323	9.025E+13
2005	1.291E+13	0.3195	2.016E+14	1.116E+13	0.2969	1.271E+14	8.65E+12	0.2612	9.361E+13	6.3E+12	0.2227	7.939E+13
2006	1.252E+13	0.3146	1.887E+14	1.092E+13	0.2937	1.245E+14	8.46E+12	0.2583	9.158E+13	6.5E+12	0.2263	8.329E+13
2007	1.278E+13	0.3179	1.972E+14	1.122E+13	0.2977	1.278E+14	7.94E+12	0.2502	8.604E+13	5.34E+12	0.2050	6.165E+13
2008	1.297E+13	0.3203	2.036E+14	1.121E+13	0.2976	1.277E+14	8.68E+12	0.2616	9.393E+13	5.91E+12	0.2157	7.199E+13
2009	1.319E+13	0.3230	2.111E+14	1.132E+13	0.2991	1.289E+14	8.47E+12	0.2584	9.169E+13	6.14E+12	0.2199	7.633E+13
2010	1.287E+13	0.3190	2.002E+14	1.059E+13	0.2892	1.208E+14	8.07E+12	0.2522	8.743E+13	5.87E+12	0.2150	7.125E+13
2011	1.268E+13	0.3167	1.939E+14	1.075E+13	0.2914	1.226E+14	7.72E+12	0.2467	8.370E+13	5.5E+12	0.2081	6.449E+13
2012	1.301E+13	0.3208	2.049E+14	1.067E+13	0.2903	1.217E+14	7.67E+12	0.2459	8.316E+13	4.72E+12	0.1927	5.105E+13
2013	1.3E+13	0.3207	2.046E+14	1.136E+13	0.2996	1.294E+14	8.13E+12	0.2532	8.807E+13	6.01E+12	0.2175	7.387E+13
2014	1.27E+13	0.3169	1.946E+14	1.103E+13	0.2952	1.257E+14	8.11E+12	0.2529	8.786E+13	6.08E+12	0.2188	7.519E+13
2015	1.247E+13	0.3140	1.870E+14	1.088E+13	0.2931	1.240E+14	8.38E+12	0.2571	9.073E+13	5.6E+12	0.2099	6.630E+13
2016	1.192E+13	0.3069	1.695E+14	1.041E+13	0.2867	1.188E+14	7.94E+12	0.2502	8.604E+13	5.37E+12	0.2056	6.218E+13
2017	1.263E+13	0.3160	1.923E+14	1.076E+13	0.2915	1.227E+14	7.94E+12	0.2502	8.604E+13	5.48E+12	0.2077	6.414E+13
2018	1.223E+13	0.3109	1.793E+14	1.078E+13	0.2918	1.229E+14	8.27E+12	0.2554	8.956E+13	5.61E+12	0.2101	6.648E+13
2019	1.219E+13	0.3104	1.780E+14	1.059E+13	0.2892	1.208E+14	7.59E+12	0.2446	8.231E+13	5.03E+12	0.1989	5.626E+13
2020	1.234E+13	0.3124	1.828E+14	1.059E+13	0.2892	1.208E+14	7.29E+12	0.2397	7.910E+13	5.07E+12	0.1997	5.694E+13
2021	1.268E+13	0.3167	1.939E+14	1.077E+13	0.2917	1.228E+14	7.65E+12	0.2456	8.295E+13	5.71E+12	0.2120	6.830E+13
2022	1.404E+13	0.3334	2.410E+14									

Arctic Sea ice

Eloise Davies

year	September			October			November			December			Yearly total of ice on orthogonal plane /m^2
	extent/m^2	ψ	monthly total of ice on orthogonal plane /m^2	extent/m^2	ψ	monthly total of ice on orthogonal plane /m^2	extent/m^2	ψ	monthly total of ice on orthogonal plane /m^2	extent/m^2	ψ	monthly total of ice on orthogonal plane /m^2	
1979	7.05E+12	0.2357	5.839E+13	8.75E+12	0.2627	1.106E+14	1.165E+13	0.3034	2.777E+13	1.367E+13	0.3289	0	1.653E+15
1980	7.67E+12	0.2459	1.027E+14	9.18E+12	0.2691	1.201E+14	1.094E+13	0.2940	1.681E+13	1.334E+13	0.3249	0	1.679E+15
1981	7.14E+12	0.2372	9.616E+13	8.86E+12	0.2644	1.130E+14	1.138E+13	0.2999	2.349E+13	1.359E+13	0.3279	0	1.657E+15
1982	7.3E+12	0.2398	9.816E+13	9.42E+12	0.2726	1.255E+14	1.093E+13	0.2938	1.666E+13	1.334E+13	0.3249	0	1.696E+15
1983	7.39E+12	0.2413	9.928E+13	9.33E+12	0.2713	1.234E+14	1.163E+13	0.3032	2.745E+13	1.364E+13	0.3285	0	1.686E+15
1984	6.81E+12	0.2316	9.201E+13	8.56E+12	0.2598	1.065E+14	1.146E+13	0.3009	2.474E+13	1.33E+13	0.3244	0	1.631E+15
1985	6.7E+12	0.2297	9.062E+13	8.55E+12	0.2597	1.063E+14	1.084E+13	0.2926	1.535E+13	1.299E+13	0.3205	0	1.640E+15
1986	7.41E+12	0.2416	9.953E+13	9.48E+12	0.2735	1.268E+14	1.102E+13	0.2950	1.800E+13	1.305E+13	0.3213	0	1.658E+15
1987	7.28E+12	0.2395	9.791E+13	9.05E+12	0.2672	1.172E+14	1.145E+13	0.3008	2.459E+13	1.322E+13	0.3234	0	1.667E+15
1988	7.37E+12	0.2410	9.903E+13	9.13E+12	0.2684	1.190E+14	1.122E+13	0.2977	2.102E+13	-9.999E+15			1.654E+15
1989	7.01E+12	0.2350	9.453E+13	8.83E+12	0.2639	1.124E+14	1.133E+13	0.2992	2.271E+13	1.363E+13	0.3284	0	1.609E+15
1990	6.14E+12	0.2199	8.348E+13	8.48E+12	0.2586	1.048E+14	1.112E+13	0.2964	1.950E+13	1.339E+13	0.3255	0	1.571E+15
1991	6.47E+12	0.2257	8.770E+13	8.54E+12	0.2595	1.061E+14	1.108E+13	0.2959	1.890E+13	1.311E+13	0.3220	0	1.597E+15
1992	7.47E+12	0.2426	1.003E+14	9.32E+12	0.2712	1.232E+14	1.088E+13	0.2931	1.593E+13	1.295E+13	0.3200	0	1.634E+15
1993	6.4E+12	0.2245	8.681E+13	8.79E+12	0.2633	1.115E+14	1.138E+13	0.2999	2.349E+13	1.341E+13	0.3257	0	1.611E+15
1994	7.14E+12	0.2372	9.616E+13	8.92E+12	0.2653	1.143E+14	1.132E+13	0.2991	2.256E+13	1.332E+13	0.3246	0	1.635E+15
1995	6.08E+12	0.2188	8.271E+13	7.83E+12	0.2484	9.136E+13	1.112E+13	0.2964	1.950E+13	1.327E+13	0.3240	0	1.524E+15
1996	7.58E+12	0.2444	1.016E+14	9.16E+12	0.2688	1.196E+14	1.076E+13	0.2915	1.419E+13	1.292E+13	0.3197	0	1.612E+15
1997	6.69E+12	0.2296	9.049E+13	8.34E+12	0.2564	1.019E+14	1.04E+13	0.2866	9.138E+12	1.286E+13	0.3189	0	1.565E+15
1998	6.54E+12	0.2270	8.859E+13	8.45E+12	0.2581	1.042E+14	1.068E+13	0.2904	1.305E+13	1.308E+13	0.3217	0	1.597E+15
1999	6.12E+12	0.2195	8.322E+13	8.6E+12	0.2604	1.074E+14	1.044E+13	0.2871	9.687E+12	1.276E+13	0.3177	0	1.590E+15
2000	6.25E+12	0.2218	8.489E+13	8.38E+12	0.2571	1.027E+14	1.08E+13	0.2921	1.477E+13	1.264E+13	0.3162	0	1.556E+15
2001	6.73E+12	0.2302	9.100E+13	8.3E+12	0.2558	1.010E+14	1.032E+13	0.2855	8.051E+12	1.264E+13	0.3162	0	1.577E+15
2002	5.83E+12	0.2142	7.949E+13	8.16E+12	0.2536	9.812E+13	1.066E+13	0.2901	1.276E+13	1.249E+13	0.3143	0	1.522E+15
2003	6.12E+12	0.2195	8.322E+13	7.85E+12	0.2488	9.176E+13	1.034E+13	0.2857	8.321E+12	1.261E+13	0.3158	0	1.529E+15
2004	5.98E+12	0.2170	8.142E+13	7.93E+12	0.2500	9.339E+13	1.013E+13	0.2828	5.518E+12	1.259E+13	0.3155	0	1.489E+15
2005	5.5E+12	0.2081	7.521E+13	7.35E+12	0.2407	8.183E+13	1.034E+13	0.2857	8.321E+12	1.255E+13	0.3150	0	1.457E+15
2006	5.86E+12	0.2148	7.988E+13	7.54E+12	0.2438	8.555E+13	1.022E+13	0.2841	6.709E+12	1.223E+13	0.3109	0	1.440E+15
2007	4.27E+12	0.1832	5.901E+13	6.04E+12	0.2181	5.779E+13	9.66E+12	0.2761	0.000E+00	1.195E+13	0.3073	0	1.372E+15
2008	4.69E+12	0.1921	6.459E+13	7.35E+12	0.2407	8.183E+13	9.76E+12	0.2775	7.940E+11	1.203E+13	0.3084	0	1.451E+15
2009	5.26E+12	0.2035	7.208E+13	6.92E+12	0.2335	7.361E+13	1.034E+13	0.2857	8.321E+12	1.236E+13	0.3126	0	1.466E+15
2010	4.87E+12	0.1957	6.696E+13	6.98E+12	0.2345	7.474E+13	9.77E+12	0.2777	9.180E+11	1.22E+13	0.3106	0	1.432E+15
2011	4.56E+12	0.1894	6.287E+13	6.46E+12	0.2256	6.517E+13	9.61E+12	0.2754	0.000E+00	1.183E+13	0.3058	0	1.379E+15
2012	3.57E+12	0.1675	4.961E+13	5.89E+12	0.2153	5.523E+13	9.77E+12	0.2777	9.180E+11	1.215E+13	0.3099	0	1.377E+15
2013	5.21E+12	0.2025	7.143E+13	7.45E+12	0.2423	8.378E+13	9.39E+12	0.2722	0.000E+00	1.201E+13	0.3081	0	1.453E+15
2014	5.22E+12	0.2027	7.156E+13	7.23E+12	0.2387	7.950E+13	9.94E+12	0.2801	3.058E+12	1.218E+13	0.3103	0	1.429E+15
2015	4.62E+12	0.1906	6.366E+13	6.97E+12	0.2343	7.455E+13	1.011E+13	0.2825	5.256E+12	1.235E+13	0.3125	0	1.390E+15
2016	4.53E+12	0.1888	6.247E+13	6.08E+12	0.2188	5.848E+13	9.85E+12	0.2788	1.918E+12	1.204E+13	0.3085	0	1.334E+15
2017	4.82E+12	0.1947	6.630E+13	6.77E+12	0.2309	7.082E+13	8.66E+12	0.2613	0.000E+00	1.146E+13	0.3009	0	1.376E+15
2018	4.79E+12	0.1941	6.591E+13	6.13E+12	0.2197	5.934E+13	9.49E+12	0.2737	0.000E+00	1.174E+13	0.3046	0	1.356E+15
2019	4.36E+12	0.1852	6.021E+13	5.73E+12	0.2124	5.255E+13	9.82E+12	0.2784	1.541E+12	1.186E+13	0.3062	0	1.327E+15
2020	4E+12	0.1773	5.540E+13	5.33E+12	0.2048	4.606E+13	9.35E+12	0.2716	0.000E+00	1.19E+13	0.3067	0	1.324E+15
2021	4.95E+12	0.1973	6.802E+13	6.82E+12	0.2318	7.174E+13	8.99E+12	0.2663	0.000E+00	1.173E+13	0.3045	0	1.391E+15
2022													1.028E+15